

Suggested Solutions to 2013 H2 Physics G.C.E. A Level Examination Paper 2

- 1 (a) Using $s = ut + \frac{1}{2}at^2$, along the direction of the motion,

Distance travelled along the direction of motion,

$$s = \frac{1}{2}(1.5)(2.0)^2 = 3.0 \text{ m}$$

Vertical height, $h = s \sin 40^\circ = 3.0 \sin 40^\circ = 1.93 \text{ m}$

- (b) (i) Using Newton's 2nd Law,

$$N - mg = ma$$

$$N = mg = 95 \times 9.81 = 932 \text{ N} \quad (\because \text{constant speed})$$

- (ii) As the cable car is moving at constant speed, there is no net acceleration, hence no net resultant force in the vertical direction. So, the normal reaction acting on the man by the floor of the cable car, N , must balance his weight, mg .

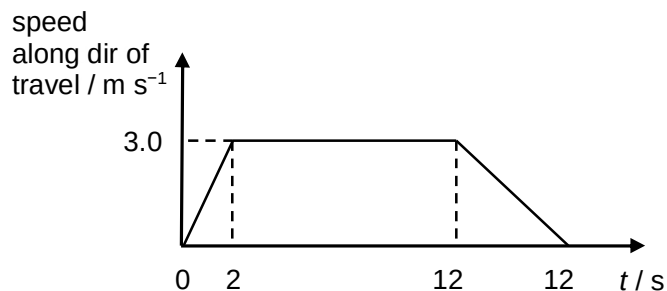
- (c) (i) Downward gravitational force.
Upward normal reaction force.
Frictional force towards the right.

Note: The existence of friction between the man and the floor of the cable car was rarely mentioned.

- (ii) Vertically, the upward normal reaction force is greater than his weight, mg , producing a resultant upward force.
Horizontally, the rightward frictional force produces a resultant rightward force.
Taken together vectorially, they produce a resultant force inclined at 40° to the horizontal, giving the man an acceleration in the direction of motion.

Note: The direction of the resultant force must be mentioned to be complete.

- (d) Graphical representation:



Distance travelled by man along direction of motion

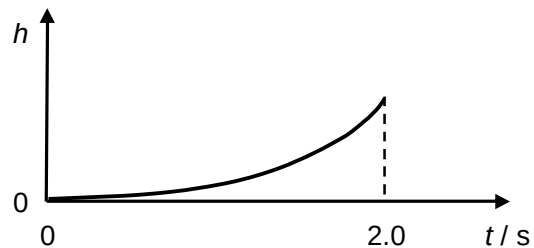
$$= \frac{1}{2}(125 + 120)(3.0) = 367.5 \text{ m}$$

Vertical height travelled by man

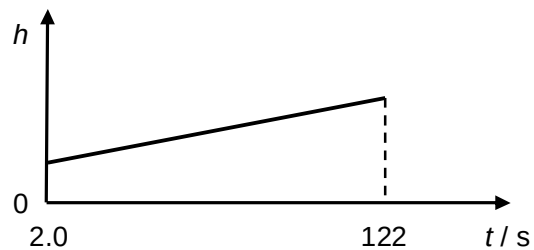
$$= 367.5 \times \sin 40^\circ = 236.2 \text{ m}$$

$$\begin{aligned}
 &\text{Gain in potential energy} \\
 &= mg\Delta h \\
 &= 95 \times 9.81 \times 236.2 \\
 &= 2.20 \times 10^5 \text{ J}
 \end{aligned}$$

(e) (i)



(ii)



(iii)

